NTIRE2026-Nighttime Image Dehazing Challenge

Zhiheng Fu
Jinan University
Guangzhou, China
fuzhiheng@stu2025.jnu.edu.cn

Shuling Zheng
Jinan University
Guangzhou, China
zhengshuling@stu2025.jnu.edu.cn

Feng Zhang
Jinan University
Guangzhou, China
zhangfeng@stu2025.jnu.edu.cn

Zhanglu Chen
Jinan University
Guangzhou, China
czhanglu@stu2024.jnu.edu.cn

Boyang Yao
Jinan University
Guangzhou, China
yaoboy@stu2024.jnu.edu.cn

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1 Team details

- Team name: JNU-620
- Team leader name: Zhiheng Fu
- Team leader address, phone number, and email: Jinan University, Guangzhou, 13822956608, fuzhiheng@stu2025.jnu.edu.cn
- Rest of the team members: Shuling Zheng, Feng Zhang, Zhanglu Chen, Boyang Yao
- Affiliation: Jinan University
- User names and entries on the NTIRE 2026 Codabench competitions: fuzhiheng
- Best scoring entries of the team during the testing phase: PSNR=25.863 LPIPS=0.1264 SSIM=0.8787
- Link to the codes/executables: `https://github.com/fuzhiheng2003/NTIRE2026_Nighttime_Image_Dehazing_Challenge.git`

JNU-620

Title: TripleSpaceDehazeNet: A Triple Color-Space Prior-Guided Network for Nighttime Image Dehazing

Members:Zhiheng Fu
Zhiheng Fu[1] *(fuzhiheng@stu2025.jnu.edu.cn)*, Shuling Zheng[1], Feng Zhang[1], Zhanglu Chen[1], Boyang Yao[1]
Affiliations:
[1] Jinan University 1

2 Contribution details

- Title of the contribution: TripleSpaceDehazeNet: A Triple Color-Space Prior-Guided Network for Nighttime Image Dehazing
- General method description:

  We propose TripleSpaceDehazeNet, a nighttime image dehazing network built on two complementary design principles. First, inspired by SGDN [1], we extend the dual color-space guidance strategy from RGB+YCbCr to a triple-space representation that additionally incorporates the CIE Lab color space.

  Second, motivated by the multi-prior guidance framework in ClearNight [2], we introduce a joint physical prior map that simultaneously encodes dark channel statistics, image gradient structure, illumination distribution, and local smoothness—four cues that together capture the heterogeneous nature of nighttime haze.

References

[1] W. Fang, J. Fan, Y. Zheng, J. Weng, Y. Tai, and J. Li, "Guided Real Image Dehazing using YCbCr Color Space," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.

[2] Y. Liu, Y. Xu, Y. Wei, X. Bi, and B. Xiao, "Clear Nights Ahead: Towards Multi-Weather Nighttime Image Restoration," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026.

- Representative image / diagram of the method(s)

  The model architecture diagram is Figure 1, at the end of the article.
- have you tested previously published methods? (yes/no) If yes, please specify which methods and the results/problems you found

  no.
- do you use extra data besides the competition data provided? (yes/no) If yes, please cite the datasets and briefly describe how did you use them (pre-processing, number of images, etc)

  no.
- Other methods and baselines tested (even if results were not top competitive)

  no.

3 Global Method Description

- Total method complexity: **Total method complexity:**
  - Parameters: 15.22 M
  - FLOPs: ≈965.42 GFLOPs per 512×512 tile
  - VRAM: ≈10 GB per batch of 2 during training
- Which pre-trained or external methods / models have been used (for any stage, if any) :

  VGG-19 (ImageNet pretrained) is used as a frozen feature extractor within the perceptual loss only. All weights of TripleSpaceDehazeNet itself are trained from scratch.
- Which additional data has been used in addition to the provided NTIRE2026-Nighttime Image Dehazing Challenge training and validation data (at any stage, if any)

  None. Only the 25 paired images from the official training split are used.
- Training description
  - Framework: PyTorch 2.8.0+cu129, NVIDIA RTX 5080 Laptop GPU
  - Training dataset: 25 paired hazy/clear images (official training split)
  - Random seed: 3407 (fixed across Python, NumPy, PyTorch CPU/CUDA; deterministic cuDNN)
  - Input resolution: 512×512 random crops (resize if image is smaller than crop size)
  - Batch size: 2
  - Optimizer: AdamW (β_1=0.9, β_2=0.999, weight decay 1×10^{-4})
  - Learning rate: 4×10^{-4} (= base $2 \times 10^{-4} \times$ batch size 2); minimum LR: 1×10^{-7}
  - LR scheduler: CosineAnnealingWarmRestarts (T_0=250, T_{mult}=1, $\eta_{\min}$=1×10^{-7})
  - Total epochs: 2000 (≈3 h on RTX 5080 Laptop)
  - Precision: Automatic Mixed Precision (AMP) with GradScaler; EMA weights maintained in float32
  - Gradient clipping: `clip_grad_norm` (max_norm= 1.0), applied after `GradScaler.unscale_()`
  - Model EMA: decay= 0.999; EMA model used for inference and checkpointing
  - Data augmentation (applied synchronously to hazy/clear pairs):
    * Random 512×512 crop (or resize if image is smaller)
    * Random horizontal flip ($p = 0.5$)
    * Random vertical flip ($p = 0.5$)
    * Random rotation from $\{0°, 90°, 180°, 270°\}$ via `TF.rotate` (uniform)
    * Mild nighttime Gaussian noise on hazy input only: level $\sim \mathcal{U}(0,\ 0.005)$, $p = 0.5$
  - Checkpoint strategy: EMA weights saved every epoch (`model_latest.pth`); full training state (model + optimizer + scheduler + scaler) saved for resumption; per-epoch snapshots saved every 20 epochs after epoch 1200

Training set usage and visualization strategy: Given the small scale of the competition training set (25 paired images), we adopt a **full-dataset training** strategy without reserving a held-out validation split. Splitting an already minimal dataset would either deprive the model of critical training samples or yield a validation set too small to produce statistically meaningful metrics. Instead, all 25 paired images participate in every training epoch, and data diversity is ensured entirely through the augmentation pipeline (random crop, flip, rotation, and mild Gaussian noise).

To compensate for the absence of a validation set and to monitor convergence qualitatively throughout training, we select a single **fixed test image depicting a complex nighttime scene**—one that simultaneously exhibits dense haze, multiple artificial light sources with visible glow halos, and regions of deep shadow—as a visual probe. Every 20 epochs, the current EMA model performs inference on this image (without augmentation), and the dehazed output is saved to disk. This longitudinal visualization sequence allows us to directly observe the progression of haze removal quality, the recovery of suppressed detail in shadow regions, and the elimination of light-source glow artifacts as training advances, providing an interpretable training signal that numerical losses on the training set alone cannot offer.

Loss function — TripleLoss: A six-component composite loss:

$$\mathcal{L} = 1.0\,\mathcal{L}_{\text{char}} + 0.005\,\mathcal{L}_{\text{fft}} + 0.1\,\mathcal{L}_{\text{percep}} + 0.5\,\mathcal{L}_{\text{color}} + 0.2\,\mathcal{L}_{\text{cos}} + 0.01\,\mathcal{L}_{\text{tv}} \tag{1}$$

Component	**Weight**	**Description**
Charbonnier (pixel)	1.0	Smooth L1, eps= 10^{-6}; applied in RGB space
FFT loss	0.005	L1 on amplitude + phase of 2D FFT
Perceptual (VGG-19)	0.1	MSE on first 12 VGG-19 layers (up to relu2_2)
YCbCr color	0.5	Charbonnier in YCbCr space; luminance + chrominance
Color cosine	0.2	$1 - \cos(\mathbf{p}, \mathbf{t})$ along channel dim; penalises hue distortion
TV on residual	0.01	TV applied to $(\hat{J} - J)$; suppresses artifacts

Table 1: TripleLoss components and weights.

Key design choices: (1) Charbonnier loss (eps= 10^{-6}) provides stable near-L1 gradients at small residuals. (2) FFT loss enforces frequency-domain consistency by penalizing both amplitude and phase. (3) YCbCr color loss separately supervises luminance and chrominance fidelity. (4) Color cosine loss penalizes hue angle distortion independently of brightness, important for preserving artificial light source colors. (5) TV loss is applied to the residual $(\hat{J} - J)$ rather than the prediction, suppressing dehazing artifacts without penalizing true image texture.

- Testing description
  - Inference: 8-way TTA (all combinations of horizontal/vertical flip and `rot90`) × overlapping tile inference (tile size 512, overlap 128, Hann-window weighted blending)
  - 8 augmented passes per image; results are inverse-transformed and averaged pixel-wise
  - Average runtime: **10.079 s / image** on NVIDIA RTX 5080 Laptop

  - Output clamped to $[0, 1]$ and saved as PNG

- Quantitative and qualitative advantages of the proposed solution
  - Triple color-space representation leverages luminance-chrominance separation (YCbCr) and perceptual lightness (Lab) for more robust dehazing under diverse nighttime illumination.
  - Joint prior map provides spatially calibrated guidance throughout the full encoder-decoder hierarchy, concentrating capacity on heavily degraded regions.
  - PGSF skip connections prevent haze-contaminated encoder features from being injected into the decoder, reducing artifact reintroduction.
  - SACEM spatially adaptive tone-curve handles the wide dynamic range between dark backgrounds and bright artificial light sources.
  - Multi-component loss jointly supervises pixel fidelity, frequency coherence, perceptual quality, color fidelity, and artifact suppression.

- Results of the comparison to other approaches (if any)

  none.

- Results on other benchmarks (if any) none.

- Novelty degree of the solution and if it has been previously published Original contribution for NTIRE2026. Novel elements include: (1) triple color-space prior bridge (RGB + YCbCr + Lab); (2) four-cue joint physical prior map; (3) PGSF skip fusion conditioned on decoder features, encoder features, and scale-matched prior; (4) spatially adaptive per-pixel tone-curve module (SACEM).

- It is OK if the proposed solution is based on other works (papers, reports, Internet sources (links), etc). It is ethically wrong and a misconduct if you are not properly giving credits and hide this information.

Please fill in the following table specifying the technical information (besides writing it) which should take 1 minute.

Ensemble refers to model ensembles and self-ensembles.

Full-resolution indicates if your model trains/infers on the original resolution of the inputs given.

Input	Training Time	Epochs	Extra data	Diffusion	Attention	Quantization	# Params. (M)	Runtime	GPU
512×512	$\approx$3 h	2000	No	No	Yes	No	15.22 M	10.079 s/img	RTX 5080 Laptop

Table 2: Technical configuration summary. Runtime is measured per test image with TTA$\times$8 and overlapping tile inference (tile 512, overlap 128).

4 Competition particularities

Any particularities of the solution for this competition in comparison to other challenges (if applicable). The primary challenge-specific adaptation is the overlapping tile inference strategy combined with 8-way TTA, designed to handle the large and variable resolution of nighttime test images while maintaining consistency with the 512×512 training crop size. Hann-window weighted blending

addresses tile-boundary stitching artifacts that are perceptually more prominent in nighttime scenes due to low average luminance. The mild nighttime noise augmentation (Gaussian, level 0–0.005) and multi-component loss design are specifically motivated by the small training set (25 images), providing implicit regularization through augmentation diversity and multi-objective supervision. Note that training rotation uses `TF.rotate` supporting $\{0°, 90°, 180°, 270°\}$, while inference TTA uses `torch.rot90` to avoid black-border artifacts on non-square test images.

5 Technical details

Please make sure to write about the language and implementation details: framework, optimizer, learning rate, GPU, datasets used for training, training time, training strategies, efficiency optimization strategies.
Implementation:

- Language: Python 3.9
- Framework: PyTorch 2.8.0, CUDA 12.9
- Optimizer: AdamW with CosineAnnealingWarmRestarts (T_0=250, $\eta_{\min}$=10^{-7})
- Hardware: NVIDIA RTX 5080 Laptop GPU
- Training dataset: 25 paired hazy/clear images (NTIRE2026 training split only)
- Training time: ≈3 h for 2000 epochs

For reproducibility, seed 3407 is applied across `random`, NumPy, and PyTorch (CPU and CUDA), with `cudnn.deterministic=True` and `cudnn.benchmark=False`. DataLoader workers are seeded via a custom `worker_init_fn`; `persistent_workers` and `pin_memory` are enabled for throughput.

6 Other details

- Planned submission of a solution(s) description paper at NTIRE 2026 workshop [YES/ NO]. NO.
- General comments and impressions of the NTIRE2026 Nighttime Image Dehazing Challenge (we appreciate your feedback to improve in future editions).
- What do you expect from a new challenge in image restoration, enhancement and manipulation?

References

[1] W. Fang, J. Fan, Y. Zheng, J. Weng, Y. Tai, and J. Li, "Guided Real Image Dehazing using YCbCr Color Space," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.

[2] Y. Liu, Y. Xu, Y. Wei, X. Bi, and B. Xiao, "Clear Nights Ahead: Towards Multi-Weather Nighttime Image Restoration," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026.

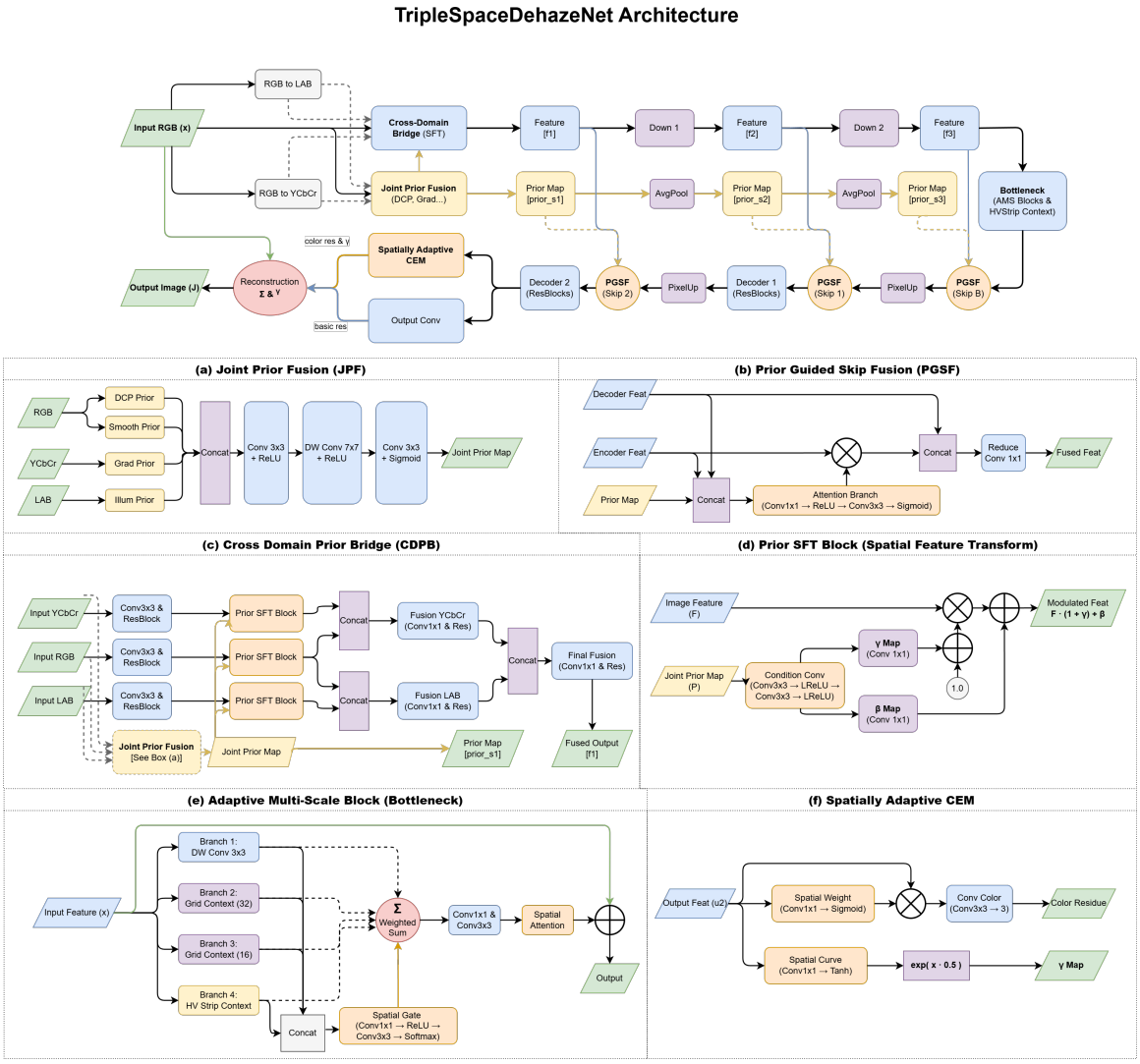

Figure 1: Architecture of TripleSpaceDehazeNet. The model builds upon a sideways U-Net structure to integrate multi-color-space features. Given the input image x, the Joint Prior Fusion (JPF) module first extracts a cross-domain physical prior map. Next, the Cross-Domain Prior Bridge (CDPB) independently encodes the RGB, LAB, and YCbCr streams. Within this process, the Prior SFT Block (Spatial Feature Transform) spatially modulates the features of each stream using the prior map, followed by cross-domain feature concatenation. At the network bottleneck, the Adaptive Multi-Scale Block (AMS) aggregates spatial and grid context features across different receptive fields. During decoding, the Prior-Guided Skip Fusion (PGSF) module introduces downsampled prior maps as attention masks. These masks filter the encoder features before concatenating them with the decoder features. Finally, the decoded features pass through the Spatially Adaptive Color Enhancement Module (CEM). This module computes spatial color residues and adaptive gamma (γ) maps to reconstruct the final image J.